

Quantum enhanced Free Energy Machine (QeFEM) algorithm

for solving NP Hard Combinatorial Optimization problems

Ronit Dutta

Singapore University of Technology and Design

July 15, 2026

Motivation

The main motivation behind this work is to ask whether optimization can be viewed as a problem in statistical physics rather than purely in computer science. Many NP-hard problems admit an Ising representation, allowing their solutions to be interpreted as ground states of interacting spin systems. The Free Energy Machine exploits this connection by replacing discrete search with variational free-energy minimization. In this talk, I'll first derive that classical framework from first principles, and then introduce a quantum extension that incorporates transverse fluctuations to improve exploration of the optimization landscape. I'll conclude by evaluating the method on standard MaxCut benchmark instances.

Discrete optimization problem

What is a Combinatorial Optimization Problem (COP)?

A problem which requires one to find an OPTIMAL OBJECT (eg: a configuration) from a set of finitely many objects satisfying a set of rules and constraints of optimality.

A combinatorial optimization problem is specified by

$$\mathbf{s} = (s_1, \dots, s_N), \quad s_i \in \{1, \dots, q\},$$

and a cost function

$$E(\mathbf{s}).$$

Optimization objective

$$\mathbf{s}^* = \arg \min_{\mathbf{s}} E(\mathbf{s}).$$

Nomenclature of COPs

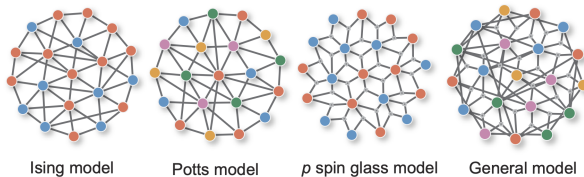


Figure 1: Types of COPs

- 1 The *QUBO/ Ising Problem* with $E(\mathbf{s}) = -\sum_{i<j} W_{ij}s_i s_j$ where s_i is a binary random variable. (eg: MaxCUT)
- 2 The *Pott's problem* allows spin states to take multivalued states i.e $s_i \in 1, 2, ..q$, yet maintaining two body interactions.
- 3 The *PUBO problem* admits p-spin interactions given by: $E(\mathbf{s}) = -\sum_{i_1 < i_2 < \dots < i_p} W_{i_1, i_2, \dots, i_p} s_{i_1} s_{i_2} \dots s_{i_p}$.
- 4 Lastly, in the *General Spin Model*, interactions can be of arbitrary order (p-spin) & spin states are discrete RVs.

Boltzmann distribution and the zero-temperature limit

At inverse temperature $\beta = 1/T$,

$$P_B(\mathbf{s}; \beta) = \frac{e^{-\beta E(\mathbf{s})}}{Z(\beta)}, \quad Z(\beta) = \sum_{\mathbf{s}} e^{-\beta E(\mathbf{s})}.$$

As $\beta \rightarrow \infty$,

$P_B(\mathbf{s}; \beta) \rightarrow$ converges to most probable ground state configuration.

Therefore,

$$\mathbf{s}^* = \arg \max_{\mathbf{s}} \lim_{\beta \rightarrow \infty} P_B(\mathbf{s}; \beta).$$

Obstacle / Primary Research Question

Accessing the Boltzmann distribution at zero would allow us to find the true distribution of states with the ground state admitting the maximum boltzmann probability (most probable). However accessing the Boltzmann distribution below critical temperature (T_c) is very difficult due to the extremely rugged landscape of spin glass systems.

Variational approximation

Choose a tractable trial distribution $P(\vec{s})$ and minimize the Kullback-Leibler divergence as

$$D_{\text{KL}}(P \parallel P_{\text{B}}) = \sum_{\vec{s}} P(\mathbf{s}) \ln \frac{P(\mathbf{s})}{P_{\text{B}}(\mathbf{s}; \beta)}.$$

Substitute the Boltzmann distribution:

$$D_{\text{KL}}(P \parallel P_{\text{B}}) = \sum_{\vec{s}} P(\mathbf{s}) \ln P(\mathbf{s}) + \beta \sum_{\mathbf{s}} P(\mathbf{s}) E(\mathbf{s}) + \ln Z.$$

Since $\ln Z$ is independent of P (constant),

$$\arg \min_P D_{\text{KL}}(P \parallel P_{\text{B}}) = \arg \min_P F[P].$$

Define

$$U[P] = \sum_{\mathbf{s}} P(\mathbf{s}) E(\mathbf{s}), \quad S[P] = - \sum_{\mathbf{s}} P(\mathbf{s}) \ln P(\mathbf{s}).$$

Variational free-energy functional

Classical/ Vanilla variational free energy

Thenrefore

$$F[P] = U[P] - \frac{1}{\beta} S[P].$$

- High temperature (or low β): entropy dominates.
- Low temperature (or high β): energy dominates.

What should be the Nature of P?

For FEM alorithm, we need to find a suitable joint distribution P. It must have:

- 1 high entropy
- 2 should be able to scalably admit q^N configurations
- 3 can solve weakly coupled, high-connectivity, or long-range limit graphs.

Variation of the Free energy function for low replica experiment

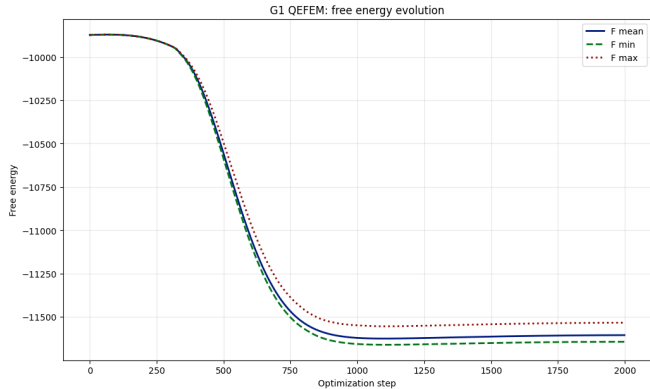


Figure 2: G4 is an 800-node GSet MaxCut graph with $\sim 1.9e4$ randomly distributed ± 1 -weighted edges. Using linear temperature and transverse annealing with only 2000 replicas, the optimizer yields 99.9742% accuracy after 2000 steps, corresponding to a gap of just 3 from the historical best 11646. Increasing the replica count subsequently closes this gap.

Variation of the Free energy function for low replica experiment

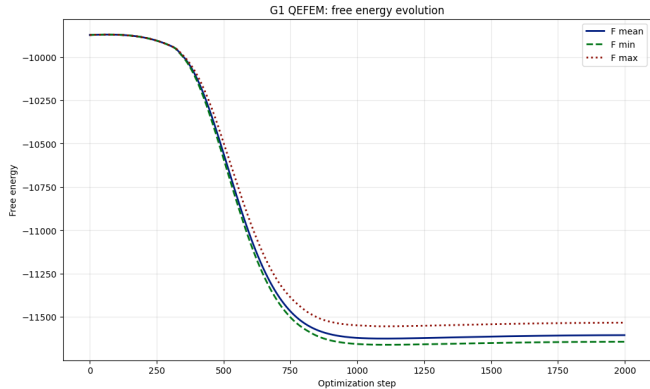


Figure 3: The real purpose of this plot is to show the behavior of free energy function over the whole run. Initially the system is unbiased, unpolarized and high temperature (entropy dominant), but with decreasing temperature the internal energy starts dominating the energy landscape after phase transition at T_c or β_c .

Entropy example from experiments

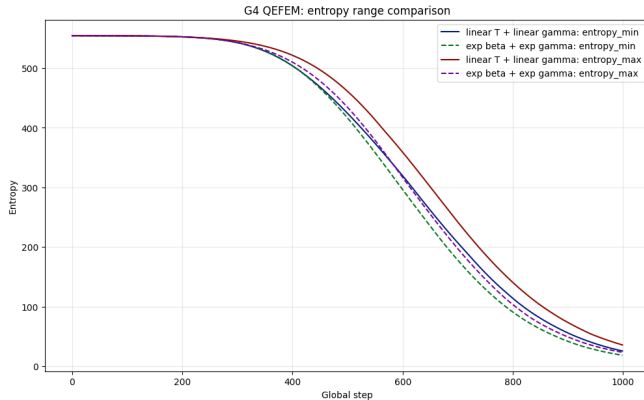


Figure 4: Behaviour of entropy under different simultaneous annealing schedules over 10^3 optimization steps using 10^4 replicas. Initially, the binary MaxCut marginals are nearly unbiased, $P_i(s_i) \approx 0.5$, corresponding to weak spin polarization and high entropy.

Entropy example from experiments

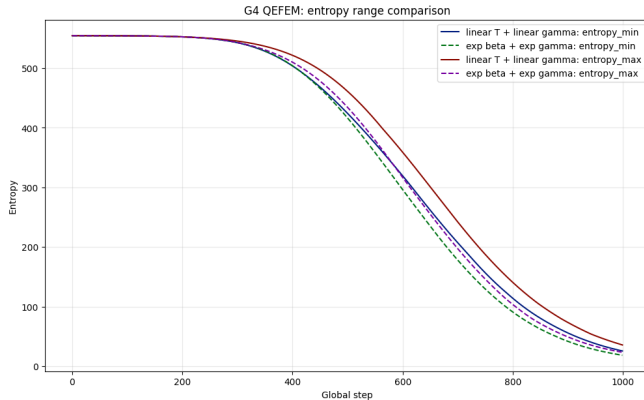


Figure 5: As β increases, entropic contribution decreases, free-energy minimization progressively polarizes the marginals and reduces the system entropy. $S_{\min}(t)$ and $S_{\max}(t)$ denote the minimum and maximum entropy across all replicas at optimization step t .

Mean-field ansatz

FEM uses a product distribution

$$P_{\text{MF}}(\sigma) = \prod_{i=1}^N P_i(\sigma_i), \quad \sum_{s=1}^q P_i(s) = 1.$$

Why use a product distribution?

- Given only single site marginals, $\prod_i P_i$ maximizes entropy with no unsupported correlations.
- It is the Gibbs state of the best non-interacting trial Hamiltonian. $H_0(\mathbf{s}) = \sum_i h_i(s_i)$
- For dense/ high connectivity and long-range systems many weak interactions self-average, so local-field fluctuations vanish and the product description becomes exact within a pure phase.

The entropy becomes exactly additive:

$$S_{\text{MF}} = - \sum_{i=1}^N \sum_{s=1}^q P_i(s) \ln P_i(s).$$

The internal energy is

$$U_{\text{MF}} = \sum_{\sigma} \left[\prod_i P_i(\sigma_i) \right] E(\sigma).$$

Unconstrained parameterization of probabilities

Mean-field free energy as a function of marginal probabilities

$$F_{\text{MF}}(\{P_i\}; \beta) = U_{\text{MF}}(\{P_i\}) - \frac{1}{\beta} S_{\text{MF}}(\{P_i\}).$$

Introduce local fields $h_i(s) \in \mathbb{R}$ which can act as continuous latent variables and define

$$P_i(s) = \frac{e^{h_i(s)}}{\sum_{s'=1}^q e^{h_i(s')}} = \text{softmax}[h_i(s_i)].$$

This guarantees that $P_i(s) \geq 0$, and $\sum_s P_i(s) = 1$.

The trainable variables are therefore $\{h_i(s)\}$ rather than $\{P_i(s)\}$.

Practical consequence

FEM turns discrete optimization into continuous optimization of the local fields.

Three qualitative stages of annealing

I. High T

$$P_i(s) \approx \frac{1}{q}$$

Entropy suppresses premature polarization.

II. Crossover

Marginals depart from uniformity and different replicas bifurcate into distinct basins.

III. Low T

Marginals become sharply peaked and the expected energy approaches a local minimum.

$$F_{\text{MF}} = U_{\text{MF}} - TS_{\text{MF}}.$$

Gradient through the softmax map

Let

$$g_i^p(s) = \frac{\partial F_{\text{MF}}}{\partial P_i(s)} \quad g_i^h(s) = \frac{\partial F}{\partial P_i(s)} \frac{\partial P_i(s)}{\partial h_i(s)} \text{ <idea behind explicit gradient>}$$

The softmax Jacobian is

$$\frac{\partial P_i(s')}{\partial h_i(s)} = P_i(s') [\delta_{ss'} - P_i(s)].$$

Hence

EXPLICIT Field gradient

$$\frac{\partial F_{\text{MF}}}{\partial h_i(s)} = P_i(s) \left[g_i^p(s) - \sum_{s'} P_i(s') g_i^p(s') \right].$$

Annealed free-energy minimization

Choose a monotone schedule

$$\beta_0 < \beta_1 < \cdots < \beta_{K-1} \implies T_0 > T_1 > \cdots > T_{K-1}.$$

At each step k :

- 1 compute $P_i(s) = \text{softmax}[h_i]_s$,
- 2 evaluate $F_{\text{MF}}(\beta_k)$,
- 3 compute gradients by autodiff or analytically,
- 4 update h using Adam, RMSprop, or SGD.

Physical interpretation

The algorithm follows a sequence of changing free-energy landscapes from **[entropy-dominated]** to **[energy-dominated]**. This transition at $T = T_c$ or alternatively $\beta = \beta_c$ creates extremely rugged energy landscape due to spin glass frustration effects.

β or Temperature annealing schedule plot

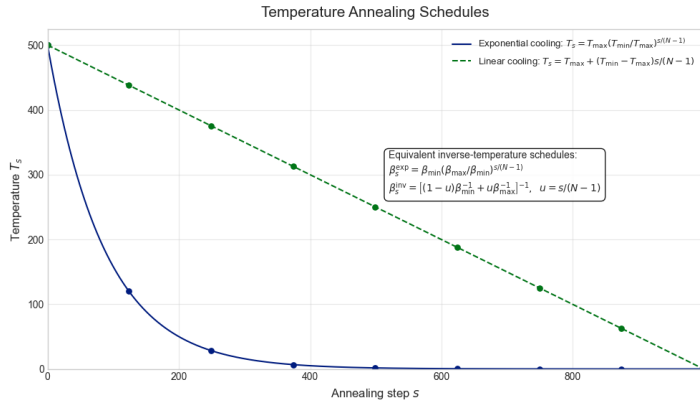


Figure 6: Temperature Annealing schedule

Why multiple replicas?

Replica Symmetry Breaking

The energy landscape of many combinatorial optimization problems is extremely rugged, containing numerous metastable minima. Spin-glass theory describes this complex organization through *replica symmetry breaking (RSB)*, motivating the use of multiple mean-field solutions to improve exploration.

Use R independently initialized mean-field distributions

$$P_{\text{MF}}^{(a)}(\sigma) = \prod_i P_i^{(a)}(\sigma_i), \quad a = 1, \dots, R.$$

Each replica explores a different basin of the non-convex free-energy landscape.

Parallel structure

- 1 All replicas can be stored in one tensor and updated simultaneously on a GPU.
- 2 At the end, search through every replica and retain the best discrete configuration.

For replica a , assign

$$\tilde{s}^{(a)} = \arg \max_s P_i^{(a)}(s_i).$$

Then evaluate the exact discrete cost

$$E\left(\tilde{s}^{(a)}\right).$$

Finally choose

$$a^* = \arg \min_a E\left(\tilde{s}^{(a)}\right).$$

Important distinction

The mean-field expected energy trains the model; the discrete cost ranks the final decoded solutions.

Why MaxCUT?

Why MaxCut?

MaxCut is a canonical NP-hard binary combinatorial optimization problem and a standard benchmark for Ising-based optimization algorithms. We evaluate QeFEM on two widely used benchmark suites: the sparse **GSet** collection and the dense **K2000** instance, comparing against the dSBM (Discrete Simulated Bifurcation Machine) baselines.

Equivalent formulations of MaxCut include:

- ① Finding the GS of an antiferromagnetic Ising Hamiltonian defined on the graph.
- ② Partitioning the vertices into two disjoint sets such that the total weight of edges crossing the partition is maximized.

Data desc

- ① The GSet graphs are a set of toroidal, random & planar sparse graphs with nodes ranging from 800-20000.
- ② The K2000 graph is a **fully-connected** graph of 2000 vertices & 1.999×10^6 edges.

MaxCUT objective

For an undirected weighted graph $G = (V, E)$ with spins $s_i \in \{-1, +1\}$,

$$C(\mathbf{s}) = \sum_{(i,j) \in E} W_{ij}(1 - \delta(s_i, s_j)) = \sum_{(i,j) \in E} W_{ij} \frac{1 - s_i s_j}{2}.$$

Maximizing the cut is equivalent to minimizing

$$E_{\text{MC}}(\mathbf{s}) = -C(\mathbf{s}).$$

Rewriting in Ising form, and removing the additive constant

$$E_{\text{MC}}(\mathbf{s}) \equiv \frac{1}{2} \sum_{(i,j) \in E} W_{ij} s_i s_j.$$

Binary mean-field variables

Define

$$p_i = P_i(s_i = +1), \quad 1 - p_i = P_i(s_i = -1).$$

The mean magnetization is

$$m_i = \mathbb{E}[s_i] = (1 \times p_i) + ((-1) \times (1 - p_i)) = 2p_i - 1$$

Under the product ansatz due to independence,

$$\mathbb{E}[s_i s_j] = \mathbb{E}[s_i] \mathbb{E}[s_j] = m_i m_j.$$

Therefore,

Expected cut

$$\mathbb{E}[C] = \sum_{(i,j) \in E} W_{ij} \frac{1 - m_i m_j}{2}.$$

Binary entropy and free energy

The single-site binary entropy is

$$S_i = -p_i \ln p_i - (1 - p_i) \ln(1 - p_i).$$

Thus

$$S_{\text{MF}} = \sum_i S_i.$$

For MaxCut,

$$F_{\text{MF}}^{\text{MC}} = - \sum_{(i,j) \in E} W_{ij} \frac{1 - m_i m_j}{2} - \frac{1}{\beta} \sum_i S_i.$$

Equivalently, ignoring the constant $-\frac{1}{2} \sum W_{ij}$,

$$F_{\text{MF}}^{\text{MC}} \equiv \frac{1}{2} \sum_{(i,j)} W_{ij} m_i m_j - \frac{1}{\beta} \sum_i S_i.$$

Classical field parameterization

For states 0, 1 the softmax equivalently reduces to sigmoid function as:

$$p_i = \sigma(h_i) = \frac{e^{h_i*1}}{e^{h_i*0} + e^{h_i*1}} = \frac{e^{h_i}}{1 + e^{h_i}} = \frac{1}{1 + e^{-h_i}}$$

Then

$$m_i = 2p_i - 1 = \tanh\left(\frac{h_i}{2}\right).$$

This representation automatically enforces

$$-1 < m_i < 1.$$

At low temperature, the marginals polarize:

$$m_i \rightarrow \pm 1,$$

and the decoded spin is

$$s_i = \text{signum}(m_i).$$

Benchmark Dataset: GSet

Dataset

- 54 weighted MaxCut instances, G1–G54.
- Graph sizes range from 800 to 3000 vertices.
- Includes random, toroidal and planar graph families.
- Sparse graphs with edge weights $W_{ij} \in \{-1, +1\}$.

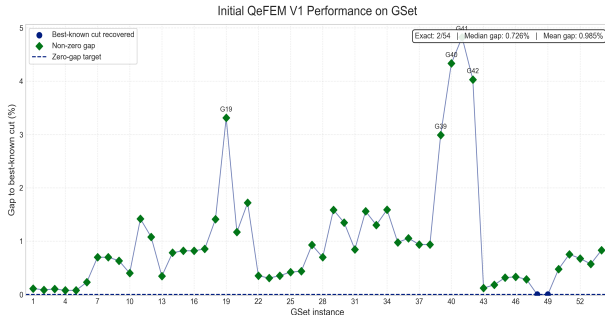
Evaluation metric

$$\text{Gap}(\%) = 100 \frac{C_{\text{BKS}} - C_{\text{QeFEM}}}{|C_{\text{BKS}}|}.$$

Gap = 0 $\rightarrow$ best-known cut recovered.

Best known cut is usually historically calculated via dSBM baselines and some other instance specific machines. They are used as an accuracy ceiling in my experiments. The current V1 experiment uses preliminary global hyperparameters and is intended primarily to check the mixed-state QeFEM implementation as a first pass across heterogeneous graph structures. Instance specific experimentation, however easy hasn't been done yet.

QeFEM V1: Overall Performance on GSet



Aggregate performance

- Improved over vanilla FEM on 54/54 instances.
- Mean gap: 5.26% $\rightarrow$ 0.98%.
- Median gap: 4.59% $\rightarrow$ 0.73%.
- Exact recovery: 2/54.

Interpretation

QeFEM V1 gives a substantial and consistent improvement in solution quality, although exact recovery remains limited under global preliminary hyperparameters.

Distribution of QeFEM V1 Solution Gaps

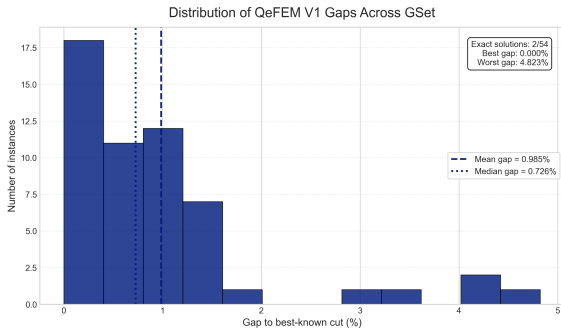


Figure 7: Aggregate distribution of gap values for naive global hyperparams

NOTE

QeFEM outperformed Vanilla FEM for the same set of hyperparameters for 54/54 GSet instances as well as for K2000!

Dense MaxCut Benchmark: K2000

Why K2000?

Unlike the sparse GSet collection, K2000 is a dense benchmark designed to evaluate optimization methods under nearly all-to-all interactions.

- Fully connected graph
- $N = 2000$ vertices
- $\approx 2.0 \times 10^6$ weighted edges
- Edge weights $W_{ij} \in \{-1, +1\}$
- Historical best cut: 33337

Computational challenge

For dense graphs, $|E| = \mathcal{O}(N^2)$, making every free-energy evaluation significantly more expensive than on sparse GSet instances.

K2000 therefore provides a complementary benchmark for evaluating scalability and optimization robustness.

Mixed-State Quantum Mean-Field Ansatz - Intro to QeFEM

Classical FEM optimizes local marginals $P_i(s_i)$. For q-states, $\hat{P}_i = \text{Diag}(p_1, p_2, \dots, p_q)$

Mixed-state QeFEM replaces each site by a single-qubit density matrix

$$\rho_i = \frac{1}{2} (\mathbb{I} + \mathbf{r}_i \cdot \boldsymbol{\sigma}), \quad \text{Tr } \rho_i = 1, \quad |\mathbf{r}_i| = R_i \leq 1.$$

Here $\mathbf{r}_i$ denotes the bloch vector for the i-th node. The global variational state is assumed to factorize,

$$\rho_{\text{MF}} = \bigotimes_{i=1}^N \rho_i.$$

Restricting to the x-z plane, $\mathbf{r}_i = R_i(\sin \theta_i, 0, \cos \theta_i)$, gives

$$\rho_i = \frac{1}{2} (\mathbb{I} + R_i \sin \theta_i X_i + R_i \cos \theta_i Z_i).$$

Quantum Mean-Field Free Energy : Single Qubit von Neumann entropy

The variational objective is

$$F_Q[\rho] = \text{Tr}(\rho H) - \frac{1}{\beta} S_{\text{vN}}(\rho),$$

For $\rho_i = \frac{1}{2} (I + \mathbf{r}_i \cdot \boldsymbol{\sigma})$, the matrix representation is

$$\rho_i = \frac{1}{2} \begin{pmatrix} 1 + r_z & r_x - ir_y \\ r_x + ir_y & 1 - r_z \end{pmatrix}.$$

Its eigenvalues follow from $\det(\rho_i - \lambda I) = 0$,
giving

$$\lambda_{\pm} = \frac{1 \pm |\mathbf{r}|}{2}, \quad \lambda_{\pm} \geq 0 \text{ (PSD matrix eigvals)}$$

Hence,

$$S(\rho_i) = -\text{Tr}(\rho_i \log \rho_i) = -\frac{1 + R_i}{2} \log \frac{1 + R_i}{2} - \frac{1 - R_i}{2} \log \frac{1 - R_i}{2}$$

Internal Energy under the Mean-Field Ansatz

The MaxCut Hamiltonian is diagonal in the computational basis,

$$H_{\text{prob}} = - \sum_{(i,j) \in E} W_{ij} \frac{I - \sigma_i^z \sigma_j^z}{2}.$$

Since $\langle \sigma_i^z \rangle = \text{Tr}(\rho_i \sigma_i^z) = r_i^z$, and $\langle \sigma_i^z \sigma_j^z \rangle = \langle \sigma_i^z \rangle \langle \sigma_j^z \rangle = r_i^z r_j^z$,
the expected cut becomes

$$\mathbb{E}[\text{Cut}] = \sum_{(i,j)} W_{ij} \frac{1 - r_i^z r_j^z}{2}.$$

Key observation

The MaxCut objective depends only on the diagonal (Z) components of the density matrix; transverse components contribute only through the quantum mixer.

Introducing transverse fluctuations

Borrowing from the TFIM hamiltonian, we allow each local states to explore 1 extra global DOF orthogonal to the computational basis by introducing a transverse field:

$$H(\Gamma) = H_{\text{prob}} - \Gamma \sum_{i=1}^N \sigma_i^x, \quad \Gamma \geq 0.$$

The local Bloch components satisfy $\langle \sigma_i^x \rangle_{\rho_i} = \text{Tr}(\rho_i \sigma_i^x) = r_i^x$.

Hence the internal-energy contribution becomes

$$U(\Gamma) = \langle H_{\text{prob}} \rangle_{\rho_{\text{MF}}} - \Gamma \sum_{i=1}^N r_i^x.$$

Role of the transverse field

The parameter Γ controls the energetic preference for transverse polarization. Large Γ favours states with nonzero r_i^x , while reducing Γ gradually restores dominance of the classical problem Hamiltonian, with $\Gamma \rightarrow 0$ basically rendering a classical solution.

Physical meaning of the QeFEM variables

$$\langle \sigma_i^x \rangle = R_i \sin \theta_i, \quad \langle \sigma_i^z \rangle = R_i \cos \theta_i.$$

- R_i controls purity / Bloch-vector length.
- θ_i controls the orientation between computational and transverse directions.
- $R_i = 0$ is maximally mixed, $R_i = 1$ is pure.

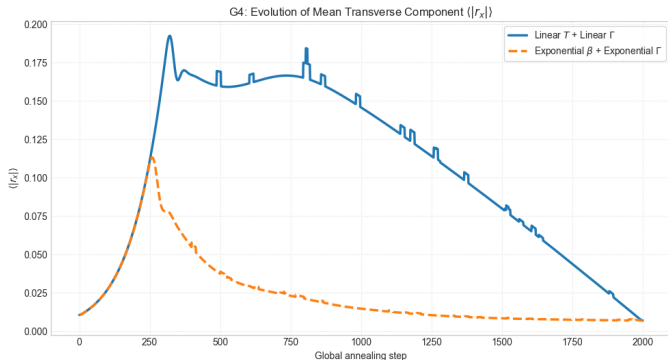
QeFEM free energy in (R_i, θ_i)

Substitute $x_i = R_i \sin \theta_i$, and $z_i = R_i \cos \theta_i$. Then

$$\begin{aligned} F_Q^{\text{MC}} = & - \sum_{(i,j) \in E} W_{ij} \frac{1 - R_i R_j \cos \theta_i \cos \theta_j}{2} - \Gamma \sum_i R_i \sin \theta_i \\ & + \sum_i \left(\frac{1 + R_i}{2} \log \frac{1 + R_i}{2} + \frac{1 - R_i}{2} \log \frac{1 - R_i}{2} \right). \end{aligned}$$

This is the continuous objective minimized during QeFEM annealing.

Evolution of Transverse Coherence



Observation

- 1 Initially, $\langle X \rangle \approx 0$, indicating negligible transverse coherence. This is because R is initialized to zero ($\sim 10^{-3}$) for maximum entropy ($-\ln 2$).
- 2 As annealing proceeds, r_x increases before gradually vanishing as $\Gamma \rightarrow 0$, allowing the optimization to transition from quantum exploration to classical refinement.

Double annealing schedule

A natural QeFEM protocol uses

$$\beta : \beta_{\min} \longrightarrow \beta_{\max}, \quad \Gamma : \Gamma_{\max} \longrightarrow 0.$$

Thermal annealing

Increasing β suppresses entropy and sharpens local density matrices.

Transverse annealing

Decreasing Γ removes the mixer and restores the classical objective.

We have used two types of annealing schedules for Transverse and Thermal annealing, namely- **Geometric** and **Linear**. For ξ denoting T or Γ , the scheduler equations will accordingly look like

$$\xi_k = \xi_{\max} \left(\frac{\xi_{\min}}{\xi_{\max}} \right)^{k/(N_{\text{steps}}-1)}, \quad \xi_k = \xi_{\max} \left(1 - \frac{k}{N_{\text{steps}} - 1} \right).$$

Transverse Annealing

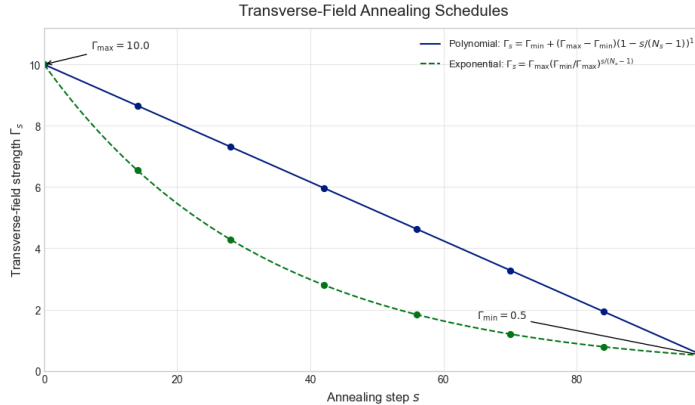
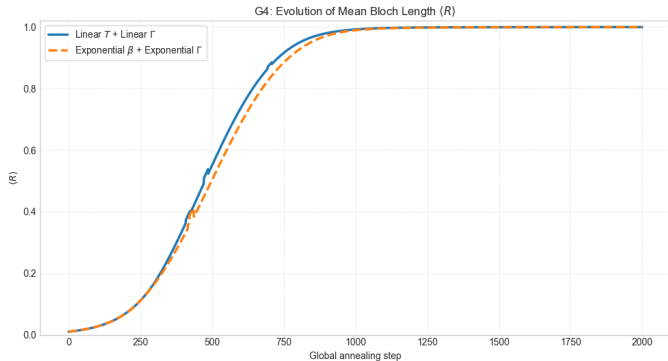


Figure 8: Transverse annealing schedules

Evolution of Purity



Bloch-vector length

- 1 The purity parameter $R = \sqrt{r_x^2 + r_z^2}$ determines the length of the Bloch vector.
- 2 The plot describes the R variation for a single instance over a full run.

Connection

Since $S_{\text{vN}} = S(R)$, the entropy evolution is completely determined by the purity.

R_x vs R_z component head to head variation

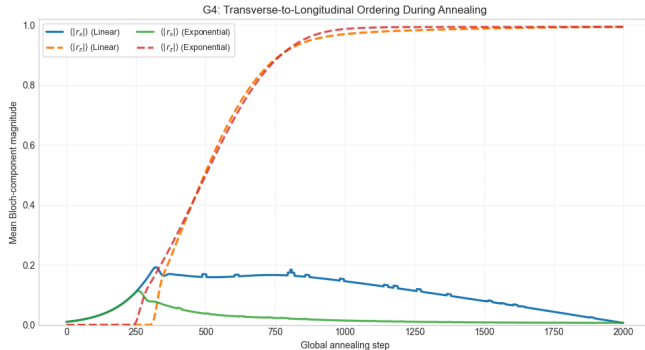


Figure 9: Variation of Transverse and Longitudinal component (aka magnetization) over 2000 N_{steps} for exponential & linear simultaneous annealing for the G4 random graph. The transverse component stays "alive" for a longer while in the linear case because the mixer schedule doesn't shrink as rapidly as exponential scheduling scheming, and gets more transverse mixing before complete polarization.

QeFEM decoding for MaxCut

The computational-basis probabilities are

$$P_i(0) = \frac{1 + r_i^z}{2}, \quad P_i(1) = \frac{1 - r_i^z}{2}.$$

Therefore decode via $s_i = \text{sign}(r_i^z) = \text{sign}(R_i \cos \theta_i) \implies r_i^z = m_i$

For each replica, the QeFEM algorithm:

- 1 decode $\mathbf{s}$ from r_i^z ,
- 2 compute the exact discrete cut,
- 3 retain the best replica.

Continuous unconstrained logits for Bloch length R

We also assume $R \geq 0$ with $R = 0$ meaning maximum entropy and vice versa meaning zero entropy pure state. For this purpose we, assign local fields h_i , just like in vanilla/classical marginals as $\{R_i\} = \text{softmax}(\{h_i\})$ and train h_i instead of R_i .

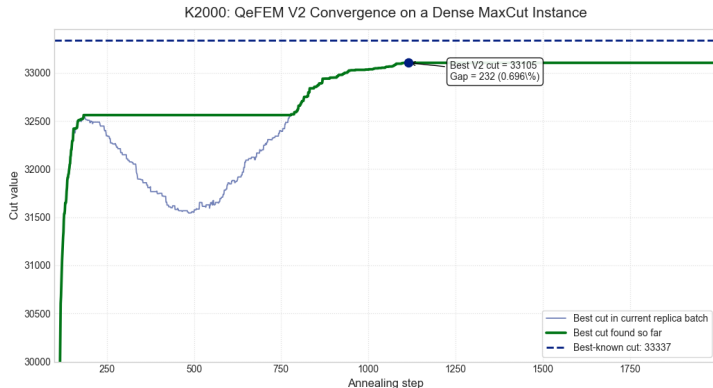
Vanilla FEM versus QeFEM

	FEM	QeFEM
Local object	$P_i(s)$	ρ_i
Ansatz	$\prod_i P_i$	$\bigotimes_i \rho_i$
Entropy	Shannon	von Neumann
Trainables	logits $h_i(s)$	h_i, θ_i
Exploration	thermal	thermal + transverse
Endpoint	discrete state	computational-basis state
Correlations	neglected	inter-site entanglement neglected

Minimal computational flow

- 1 Read graph and construct W (adjacency matrix via dense loader or edge-list).
- 2 Initialize R replicas near the high-entropy state.
- 3 Construct schedules β_k and Γ_k .
- 4 Evaluate F_{MF} or F_Q^{MF} .
- 5 Differentiate Free energy function using automatic differentiation (Adam, SGD, RMSProp) or via explicit chain-rule formulas.
- 6 Update trainable parameters.
- 7 Decode every replica to find discrete cut and hard cut value.
- 8 Evaluate exact discrete costs by decoding the spins.
- 9 Across replicas report best cut, gap, ecut, entropy, and accuracy percentage, and ever successes (gap=0).

QeFEM V2 Performance on Dense K2000



V2 result

$$\begin{aligned}C_{\text{BKS}} &= 33337, \\C_{\text{V2}} &= 33105, \\ \text{Gap} &= 232, \\ \text{Gap}(\%) &= 0.696\%.\end{aligned}$$

QeFEM Convergence on Large GSet Instances

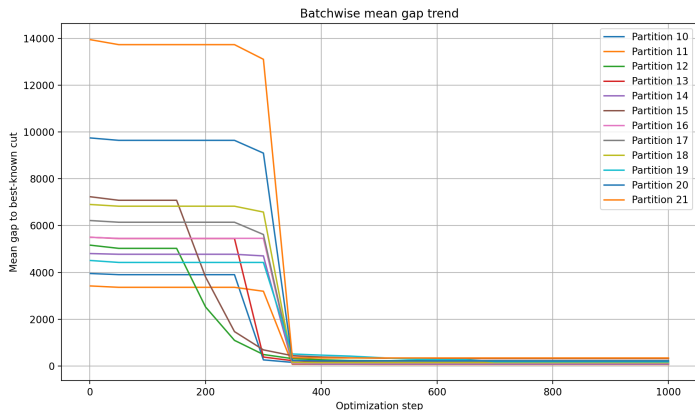


Figure 10: Each curve reports the mean gap within one computational batch.

Batch composition

10	G55, G56	16	G65
11	G57	17	G66
12	G58, G59	18	G67, G72
13	G60, G61	19	G70
14	G62	20	G77
15	G63, G64	21	G81

Why separate these batches?

Partitions 10–21 comprise the largest GSet benchmarks (G55–G81), containing graphs from 5k to 20k vertices. Owing to their substantially larger problem sizes, they are analyzed independently from the G1–G54 benchmark suite.

v1 run: r_x variation for Big GSet graphs

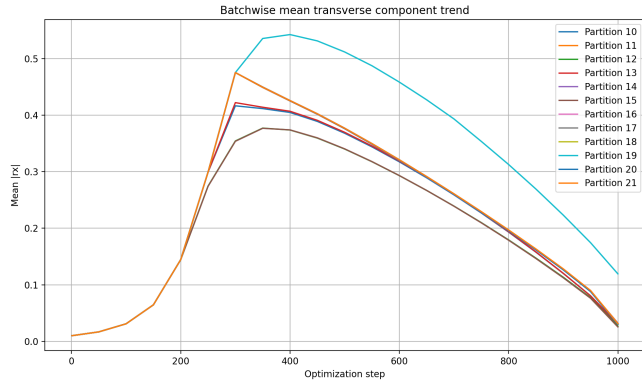


Figure 11: Transverse component (mean absolute $|r_x|$) variation for big gset partitions

Core hyperparameters

Category	Hyperparameters
Annealing	$\beta_{\min}$, $\beta_{\max}$, schedule type, N_{step}
Quantum mixer	$\Gamma_{\max}$, $\Gamma_{\min}$, schedule type
Parallel search	number of replicas R
Optimizer	optimizer type, learning rate, momentum / moments
Regularization	weight decay, gradient scale, clipping
Initialization	field scale; R_i and θ_i initialization
Numerics	dtype, device, stabilizing ϵ

Time-to-solution metrics & supposed "Quantum Advantage"

Time efficiency metrics like TTS and TTS99 have not been showed in this presentation because I have yet not conclusively run per instance $gap = 0$ runs, upon which the hyperparams shall be pruned, and tweaked and trimmed to get the optimal values. That would take some time and for that reason avoided.

Appendix: q -state/ qudit based QeFEM - Beyond Binary QeFEM

Classical FEM naturally generalizes from the Ising model ($q = 2$) to the q -state Potts model. The corresponding mixed-state quantum formulation replaces qubits by q -level qudits.

Generalized local density operator

For a q -state system,

$$\rho_i = \frac{1}{q} \left(I + \sum_{a=1}^{q^2-1} r_i^{(a)} \Lambda_a \right),$$

where $\{\Lambda_a\}_{a=1}^{q^2-1}$ are the generators of the Lie algebra $SU(q)$.

Generalized Pauli formalism

The Pauli matrices form the generators of $SU(2)$. For higher-dimensional qudits, they are replaced by the generalized Gell–Mann matrices (or any equivalent $SU(q)$ generator basis), providing a complete Hermitian basis for mixed-state tomography.

Appendix: Potts's QeFEM Algorithmic Scaling

Parameter scaling

- Classical Potts FEM requires approximately $\mathcal{O}(NRq)$ trainable marginal parameters.
- The mixed-state quantum formulation requires one generalized Bloch vector per site, $q^2 - 1$ real coefficients, giving $\mathcal{O}(NR(q^2 - 1))$ trainable variables.
- The initialization and the properties of the single qudit density matrix pose real future research scope, given q-state qudits.

Research challenges

- Initialization of generalized Bloch vectors/ density matrices.
- Increased computational and memory complexity.
- Investigation of improved accuracy and reduced hyperparameter sensitivity.
- Possible improvement of TTS metrics compared to Vanilla Pott's and QUBO.

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Thank you